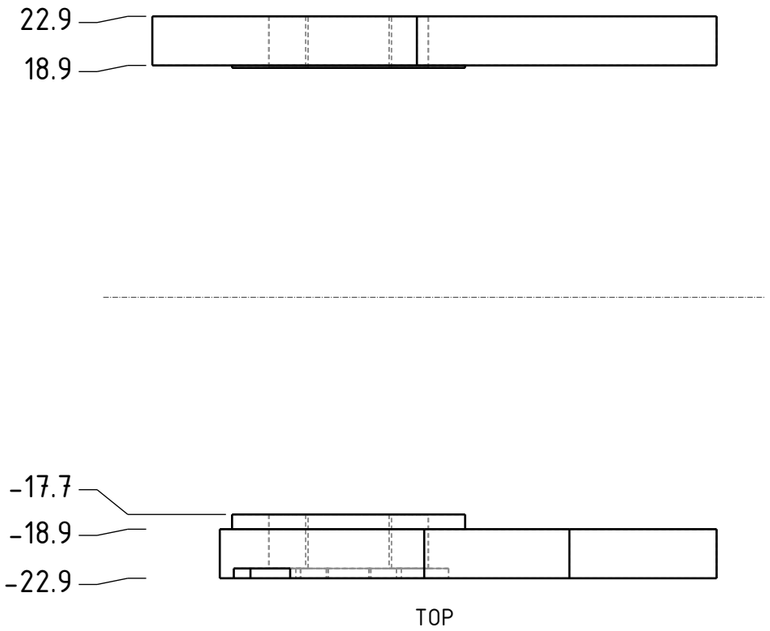
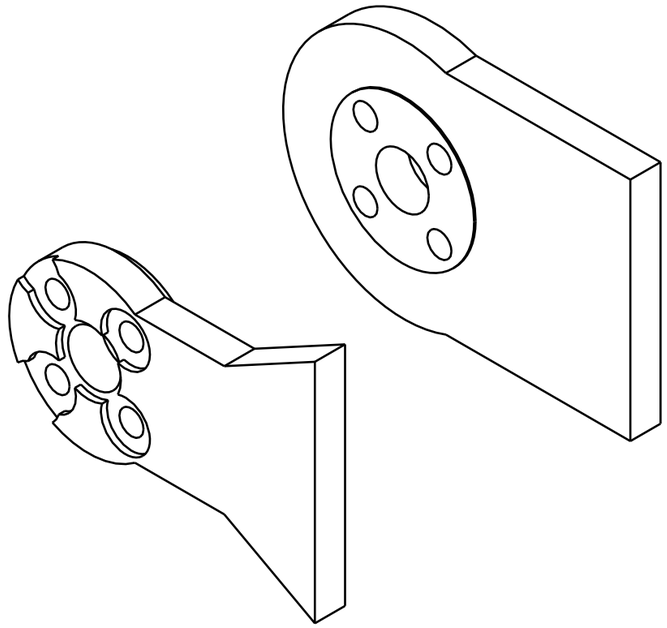


TOP



ISOMETRIC (NOT TO SCALE)

IDLER NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	0.6R = 18	±8.5
N4	R = 30	±H = ±15

CLEVIS (every pitch joint)

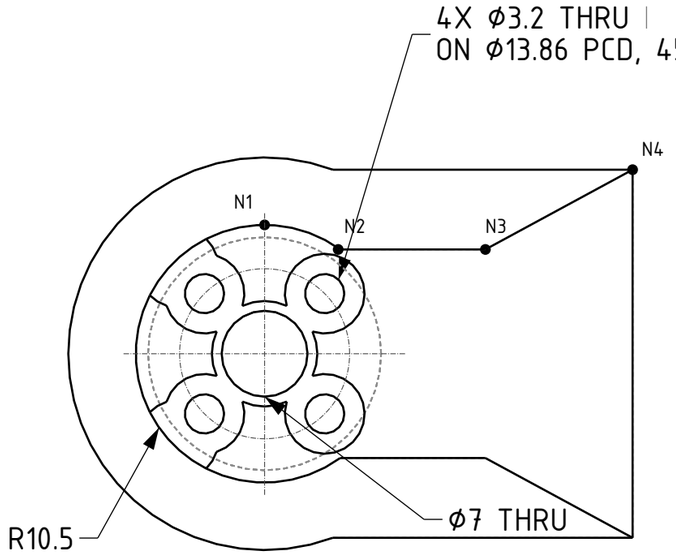
Drawn for the arm links: reach R = 30, half-height H = 15. The wrist uses R = 24, H = 12.

Not printed on its own: the plates are modelled into the links. On TAL-103 and TAL-104 the idler plate is the separate part TAL-110; on the wrist (TAL-105) both plates are wrist body.

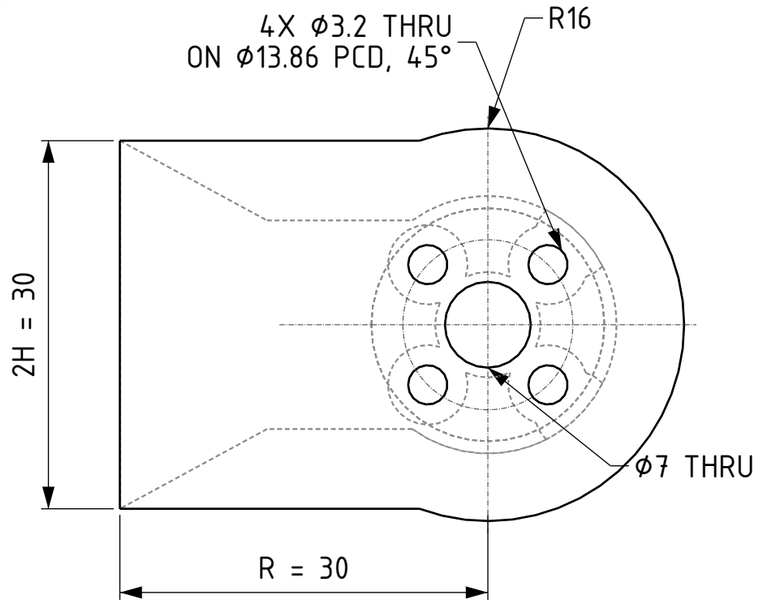
1. Output plate (+Y), y 18.9 to 22.9: circle R16 at the origin + rectangle x 0 to R, z ±H.
2. Output plate pad ⌀19 on the inner face, y 18.7 to 18.9: bears on the output horn face (the plate itself clears the parent servo's shell by 0.5).
3. Output plate holes, through: ⌀7 centre (horn spline screw) + 4X ⌀3.2 on ⌀13.86 PCD at 45°.
4. Idler plate (-Y), y -22.9 to -18.9: circle R10.5 + neck polygon N1 to N4 (mirrored). The neck clears the parent servo's bus plugs.
5. Idler plate pad ⌀19 on the inner face, y -18.9 to -17.7, bearing on the passive horn; centre ⌀7 through (clears the idler boss); 4X ⌀3.2 on ⌀13.86 PCD at 45°, counterbore ⌀6.5 0.8 deep from outside (4X M3 × 6 into the passive horn, which turns on the boss); ⌀8.6 0.8 deep on the centre, and the two counterbores at x < 0 run out through the hub rim (see TAL-110).
6. Clevis gap in the link body: remove a cylinder R21 about Y between y ±18.9, and everything at x < 0 between those planes, except the two horn pads (dimensioned on each link's section).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.



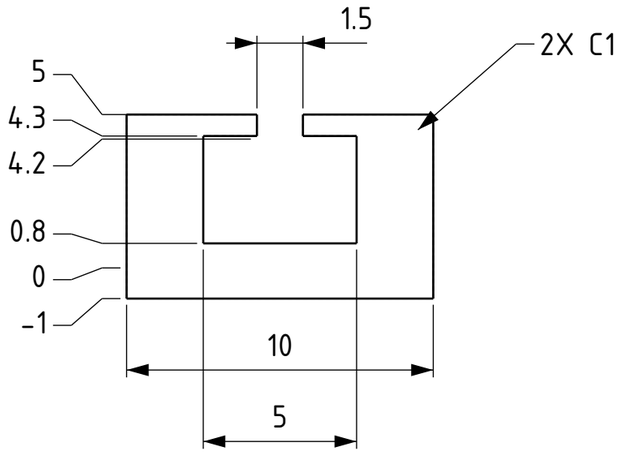
FRONT (IDLER PLATE, -Y)



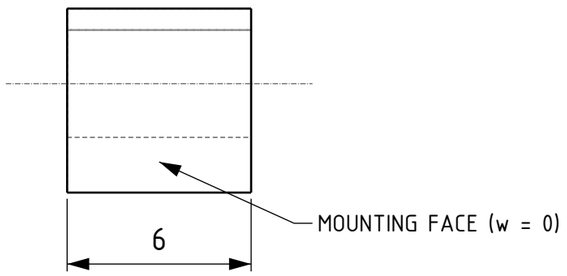
REAR (OUTPUT PLATE, +Y)

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clevis (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-07
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 21



CABLE CLIP: END VIEW



CABLE CLIP: SIDE VIEW

SEAT S (FROM THE SERVO MID-PLANE)

PART	S
arm shells, gripper	17.3
shoulder	18.5
base	18.4
wrist	18.0

COMMON FEATURES

CABLE CLIP, one bus cable. Local u along the cable, v across, w out of the mounting face.

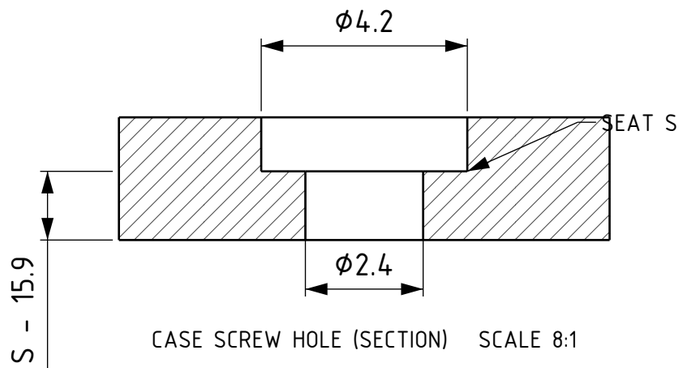
1. Block u ±3, v ±5, w -1 to 5 (sunk 1 into the face).
2. Tunnel through along u: v ±2.5, w 0.8 to 4.3. Slit v ±0.75 from w 4.2 out through the top.
3. Chamfer C1 on the two outer long edges.

CASE SCREW HOLE (M2 self-tap into the servo case): ⌀2.4 from the servo face to the seat S, then ⌀4.2 counterbore from S outward through the part.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa, zc) (xa+k, zc-k) (xb-k, zc-k) (xb, zc) (xb-k, zc+k) (xa+k, zc+k), cut through the part along Y. Drawn for H(0, 40, 0, 10).

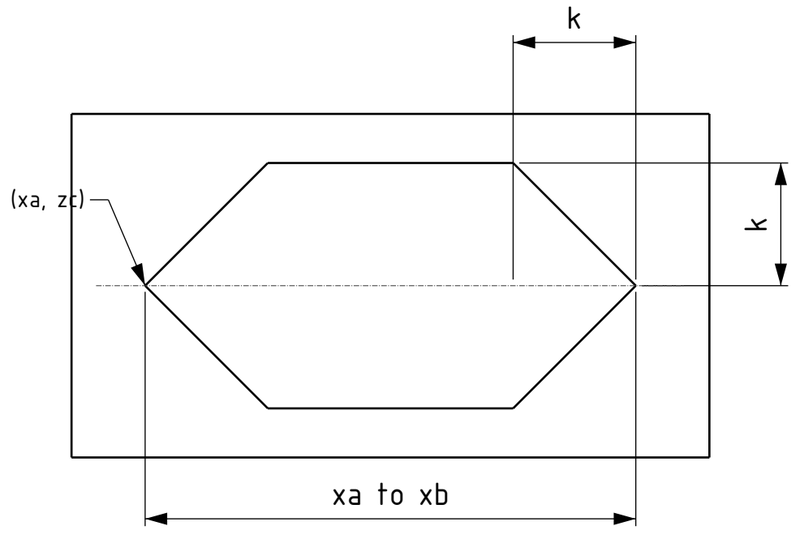
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



CASE SCREW HOLE (SECTION) SCALE 8:1

SERVO FACE (BOTTOM) IS 15.9
FROM THE SERVO MID-PLANE



HEX WINDOW H(xa, xb, zc, k) SCALE 2:1

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clip, screw hole, window (part sheet 2 of 2)	SCALE 5:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-07
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 21